High-frequency data and limit order books
April 15th, 2025

Length : 2h00. This exam paper is three page-long. The four exercises are independent.
Documents and electronic devices are NOT allowed.
You may answer in either French or English. Vous pouvez répondre en français ou en anglais.

Exercise 1 - Asymptotic properties of Poisson processes

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space. Let $(N_t)_{t\geq 0}$ a time-homogeneous Poisson process with intensity $\mu > 0$.

1. Prove that $\frac{N_t}{t} \xrightarrow[t\to+\infty]{L^2} \mu$ $(t \in \mathbb{R}_+)$.
2. Prove that $\frac{N_n}{n} \xrightarrow[n\to+\infty]{} \mu$ $\mathbf{P}$-a.s. $(n \in \mathbb{N})$.
3. Prove that $\frac{N_t - \mu t}{\sqrt{\mu t}} \xrightarrow[t\to+\infty]{} \mathcal{N}(0,1)$ in distribution $(t \in \mathbb{R}_+)$, where $\mathcal{N}(0,1)$ denotes the standard Gaussian distribution.

Exercise 2 - Realized variance

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbf{P})$ be a filtered probability space. Let $(W_t)_{t\geq 0}$ be a $(\mathcal{F}_t)$-Brownian motion. Let $\sigma > 0$ and let $p_t = \sigma W_t$ represent the log-price of a financial asset. Let $T > 0$ be a finite horizon (assume for simplicity that $T \in \mathbb{N}$). For a given $n \in \mathbb{N}^*$, let $t_i^{(n)} = \frac{i}{n} = i\frac{T}{nT}$, $i = 0, \ldots, nT$. Let $X_{n,j} = p(t_j^{(n)}) - p(t_{j-1}^{(n)})$, $j = 1, \ldots, nT$. Let θ_n be the standard realized variance estimator computed for the process p on $[0,T]$ with the subdivision $0 = t_0^{(n)} < t_1^{(n)} < \ldots < t_{nT}^{(n)} = T$.

1. What is the financial interpretation of the $X_{n,j}$'s ? Justify.
2. State the mathematical definition of θ_n.
3. Compute $\mathbf{E}[\theta_n]$ and $\mathbf{V}[\theta_n]$. Recall that the fourth moment of the standard Gaussian distribution is equal to 3.
4. What quantity does θ_n estimate (mathematically and financially) ? Show that the estimation is consistent.
5. Prove that the estimator is asymptotically Gaussian. (It may be useful to consider $n\theta_n$.)
6. State without proof the analogous result(s) in the general case where $p_t = \int_0^T \sigma_s dW_s$ for some stochastic volatility process $(\sigma_s)_{s\geq 0}$.
7. What is the *volatility signature plot* in high-frequency finance ? Draw its expected shape for equity data.

Exercise 3 - Hawkes processes

Let $(N_t)_{t\geq 0}$ be a unidimensional Hawkes process with constant baseline intensity $\mu > 0$ and excitation kernel $h(t) = \alpha e^{-\beta t}$, $t \geq 0$, for some constants $0 < \alpha < \beta$. We consider the standard case discussed in the lectures : the Hawkes process is causal, defined on $[0, +\infty)$, i.e. there are no events on $(-\infty, 0)$.

1. State the expression of the conditional intensity $\lambda(t)$ of the process N.
2. Let $\varphi(t) = \mathbf{E}[\lambda(t)]$, $t \in [0, +\infty)$. Compute the Laplace transform $\tilde{\varphi}(s) = \displaystyle\int_0^{+\infty} e^{-st}\varphi(t)\,dt$, $s > 0$.
3. Show that $\tilde{\varphi}(s) = \dfrac{\mu}{\beta - \alpha}\left(\dfrac{\beta}{s} - \dfrac{\alpha}{s + \beta - \alpha}\right)$.
4. Invert the Laplace transform to obtain φ (if necessary, quickly compute the Laplace transforms of a constant function and of an exponential function).
5. Compute $\lim\limits_{t \to +\infty} \varphi(t)$. Is this result in agreement with your knowledge of Hawkes processes ? State (without proof) the result in the case of a general kernel $\nu : [0, +\infty) \to [0, +\infty)$.

Exercise 4 - Miscellaneous questions

1. **On high-frequency datasets.** The following images are extracts from a dataset similar to the one you worked with during the labs.

ets	etype	eprice	eqty	eside	bp0	bq0	ap0	aq0
2017-01-19 10:01:50.384622	C	59610	259	S	59580	1050	59590	424
2017-01-19 10:01:50.897262	T	59590	424	S	59580	1050	59600	1468
2017-01-19 10:01:50.897281	C	59610	200	S	59580	1050	59600	1468
2017-01-19 10:01:50.897287	A	59580	200	B	59580	1250	59600	1468
2017-01-19 10:01:50.897296	C	59610	200	S	59580	1250	59600	1468
2017-01-19 10:01:50.897489	A	59580	42	B	59580	1292	59600	1468

ets	etype	eprice	eqty	eside	bp0	bq0	ap0	aq0
2017-01-19 15:41:55.556798	A	46005	200	S	45985	348	46005	1855
2017-01-19 15:41:55.560494	A	46000	250	S	45985	348	46000	250
2017-01-19 15:41:55.560911	C	46000	250	S	45985	348	46005	1855
2017-01-19 15:41:55.560917	A	46000	189	S	45985	348	46000	189

ets	etype	eprice	eqty	eside	bp0	bq0	ap0	aq0
2017-01-27 10:02:57.453311	A	61150	200	S	61120	1387	61140	218
2017-01-27 10:02:57.471396	M	61140	6	S	61120	1387	61140	364
2017-01-27 10:02:57.473611	C	61120	200	B	61120	1187	61140	364
2017-01-27 10:02:57.476322	C	61120	200	B	61120	987	61140	364

ets	etype	eprice	eqty	eside	bp0	bq0	ap0	aq0
2017-01-19 15:35:31.113126	C	46105	200	S	46090	663	46100	163
2017-01-19 15:35:31.113216	A	46100	163	B	46090	663	46100	163
2017-01-19 15:35:31.113219	T	46100	163	S	46090	663	46105	1009
2017-01-19 15:35:31.113353	A	46085	188	B	46090	663	46105	1009

   (a) State a *precise* definition of each column of the dataset.
   (b) For each of the 4 extracts, describe the events occuring at lines 2 and 3 and their impact on the limit order book.
   (c) Are there any incoherences in these extracts ?
   (d) Sign each of these events with respect to the pressure it exerts on the mid-price. Relate your answer to a well-known stylized fact of high-frequency financial markets.
   (e) For all transactions of each extract, is the observed sign of the trade the most probable one given the imbalance ?

2. **On the Epps effect.**
   (a) What is known as the Epps effect in finance ?

(b) State the definition of the Hayashi-Yoshida estimator.

(c) What is the interest of the Hayashi-Yoshida estimator in high-frequency finance ?

3. **On propagator models.** Let us consider a mid-price model in trade time with propagator $G : \mathbb{N} \to \mathbb{R}$. Let $(p_m)_{m\in\mathbb{Z}}$ be the sequence of mid-prices. Let $(\epsilon_m)_{m\in\mathbb{Z}}$ be the sequence of the signs of the trades. Let $(\eta_m)_{m\in\mathbb{Z}}$ be a sequence of i.i.d. random variables such that $\mathbf{E}[\eta_0] = 0$ and $\mathbf{V}[\eta_0] = \sigma^2 < \infty$. The mid-price at time n (just before the n-th trade) is

$$p_n = \sum_{m<n} \left[G(n-m)\epsilon_m + \eta_m\right].$$

   (a) Compute $\mathcal{R}(\ell) = \mathbf{E}[(p_{n+\ell} - p_n)\epsilon_n]$, $n, \ell \in \mathbb{N}$. How is the function $\mathcal{R}$ called and what does it represent ?

   (b) Compute $\mathcal{R}(\ell)$ when G is constant and $(\epsilon_m)_{m\in\mathbb{Z}}$ are i.i.d. random variables. Explain the result.

   (c) We still assume that G is constant but $(\epsilon_m)_{m\in\mathbb{Z}}$ are no longer i.i.d. random variables. Let $r_n = p_{n+1} - p_n$. Compute $\mathbf{E}[r_{n+\ell}r_n]$. What is this quantity ? Why is the result problematic ?

4. **On optimal execution.**

   (a) Explain the concept(s) of optimal execution on high-frequency financial markets.

   (b) Describe briefly the assumptions and findings of the Obizhaeva & Wang model. No mathematical proofs or calculatory developments are expected.

••• Fin de l'énoncé. •••